

**Hybrid Cellular Automata - Random Forest Framework Scenario Based  
Simulation of Future Urban Dynamics Around the Proposed International  
Greenfield Airport at Parandur, Chennai**

# INTRODUCTION

- Urbanization is one of the major drivers of land use and land cover change, particularly around rapidly developing metropolitan regions.(Gumma et al.,2017)
- Large-scale infrastructure projects, such as airports, often accelerate urban expansion by attracting residential, commercial, industrial, and transportation developments. (Swangjang & Iamaram, 2011)
- While such growth supports economic development, it can also lead to the loss of agricultural land, environmental degradation, and unplanned urban sprawl if not managed effectively.(Gumma et al.,2017)
- The Proposed International Greenfield Airport at Parandur, located near Chennai in the boundary of CMA, is expected to become a major growth center and significantly influence future urban development in the surrounding region. (Government of Tamil Nadu, 2022; Chennai Metropolitan Development Authority, 2024).

# INTRODUCTION (*contd.*)

- The development of major transportation infrastructure has historically altered land-use patterns by improving accessibility, increasing land values, and attracting investments in residential, commercial, industrial, and institutional developments. Consequently, areas surrounding airports often experience urbanization compared to other peri-urban regions (Kasarda & Lindsay, 2011).
- Scenario-based urban growth modelling provides a systematic approach to assess how different planning and development policies may shape future land-use patterns.
- Among the available urban growth models, the hybrid Cellular Automata–Random Forest (CA–RF) framework has gained increasing attention because it combines the high predictive capability of Random Forest with the spatial simulation strength of Cellular Automata.
- Therefore, this study aims to simulate future urban dynamics around the Proposed International Greenfield Airport at Parandur under multiple development scenarios using a hybrid CA–RF framework integrated with Remote Sensing and GIS.

# REVIEW OF LITERATURE

- The Land use patterns have been noted to have changed before the commencement of mega project construction. The land use pattern around the proximity of the proposed project area experienced on sharp incline toward urbanization (Swangjang & Iamaram, 2011)
- There are negative effects of aviation noise and pollution on residential or other related construction. Aviation noise and pollution have the greatest impact on the environment. Which maintain a close relationship with living or working (Xiong et al., 2018)
- In India four cities (Ahmedabad, Bengaluru, Chennai, and Hyderabad) are currently having 5-10 million inhabitants, projected to become megacities (population of  $> 10$  million) in the coming year with the total of seven megacities projected by the country by 2030. (United Nation, 2014)

# REVIEW OF LITERATURE (*contd.*)

- Managing urban expansion in future is a important thing for agricultural growth and food security. Also providing the common amenities such as housing, water, transportation and employment for the growing population (Gumma et al., 2017)
- Chennai has experienced rapid urbanization and industrial expansion, particularly in the manufacturing, information technology, and construction sectors, which have attracted both skilled professionals and unskilled migrant workers from different parts of India, thereby accelerating urban growth and increasing demand for housing and infrastructure (Krishnamurthy & Desouza, 2015; Rao, 2013).
- The Chennai Metropolitan Area (CMA) was initially delineated in 1974 with an extent of 1,189 km<sup>2</sup>. To accommodate rapid urbanization and promote integrated regional planning, the Government of Tamil Nadu expanded the CMA boundary to 5,904 km<sup>2</sup> in 2022 through G.O. Ms. No. 184, incorporating 1,225 additional villages from the surrounding districts (Government of Tamil Nadu, 2022; CMDA, 2024).

# REVIEW OF LITERATURE (*contd.*)

- Using Landsat satellite imagery, Random Forest (RF), and Cellular Automata–Markov (CA–Markov) models, future land use and land cover (LULC) changes can be effectively simulated by integrating machine learning with spatial modelling techniques. (Badshah et al., 2024).
- Future land-use simulation under multiple development scenarios is essential for supporting sustainable urban planning and decision-making. Liu et al. (2021) developed a **Cellular Automata–Random Forest (CA–RF)** model to predict intra-urban land-use changes in Kumamoto City, Japan, under two scenarios: an unconstrained development scenario and a planning-constrained scenario that incorporated disaster-related factors. (Liu et al., 2021)
- Airport infrastructure has the potential to significantly reshape surrounding landscapes by accelerating urban development and altering land-use patterns. The integration of Remote Sensing, GIS, and spatial simulation models provides an effective framework for assessing the future impacts of airport-induced urbanization and supporting sustainable regional planning (Kasarda & Lindsay, 2011).

# RESEARCH GAP

- Most previous studies focused on existing urban areas or operational airports, whereas limited studies have investigated future urban growth around a proposed Greenfield airport before its development.
- Airport-related studies mainly assessed land-use changes after airport construction, but they did not evaluate future urban expansion under multiple infrastructure development scenarios.
- Many studies used CA–Markov, Random Forest, or other individual models, whereas few studies have integrated a **Hybrid Cellular Automata–Random Forest (CA–RF)** approach for scenario-based urban growth prediction.
- Previous studies primarily analysed historical LULC changes, with limited emphasis on predicting future urban dynamics using machine learning integrated with spatial simulation.

## RESEARCH GAP (*contd.*)

- Most scenario-based studies considered only one or two scenarios (e.g., unrestricted vs. planning-constrained development), while comparative analysis of Business-as-Usual, Airport Development, Metro Connectivity, and Airport + Metro Connectivity scenarios is lacking.
- There is a lack of studies addressing the combined influence of major transportation infrastructure (airport and metro) on future urban growth in rapidly developing metropolitan regions.
- No published study has assessed future urban dynamics around the Proposed International Greenfield Airport at Parandur using Remote Sensing, GIS, and a Hybrid CA–RF workflow.

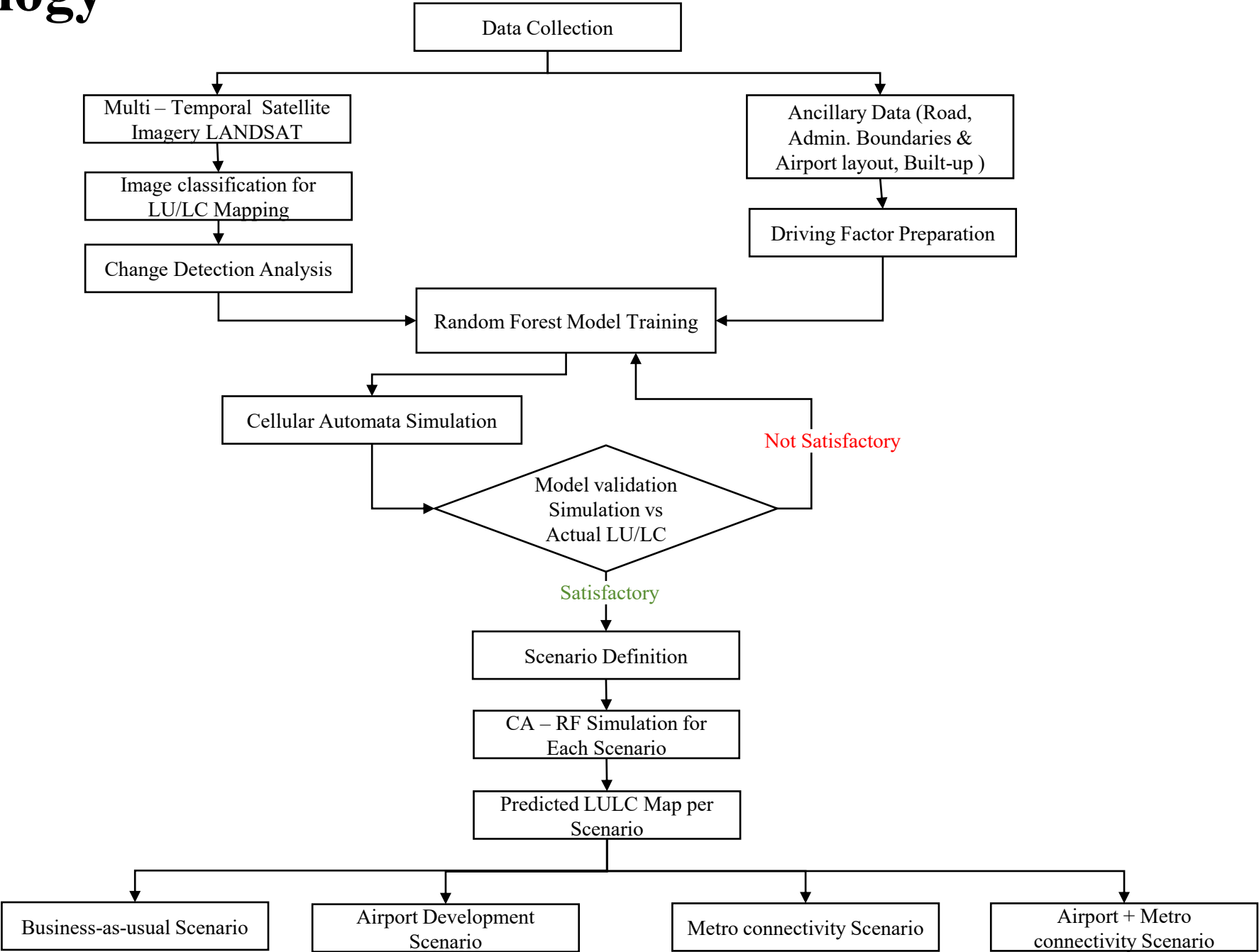
# AIM

This project aimed to simulate future urban dynamics around the Proposed International Greenfield Airport at Parandur using a hybrid Cellular Automata-Random Forest (CA-RF) framework integrated with Remote Sensing and GIS.

## OBJECTIVES

1. Preparation of Land Use/Land Cover (LULC) maps (2010, 2015, 2020 & 2025) of the study area using temporal satellite imageries.
2. Identification of key factors influencing urban growth, such as roads, existing built-up areas, and the proposed airport.
3. Development of hybrid CA-Random Forest model for simulating future urban expansion,
4. Comparison of predicted LU/LC for 4 different scenarios such as :
  - Business-as-Usual,
  - Airport Development,
  - Metro Connectivity, and
  - Airport + Metro Connectivity and
5. Analysis of impacts of the proposed airport on future urban expansion patterns.

# Methodology

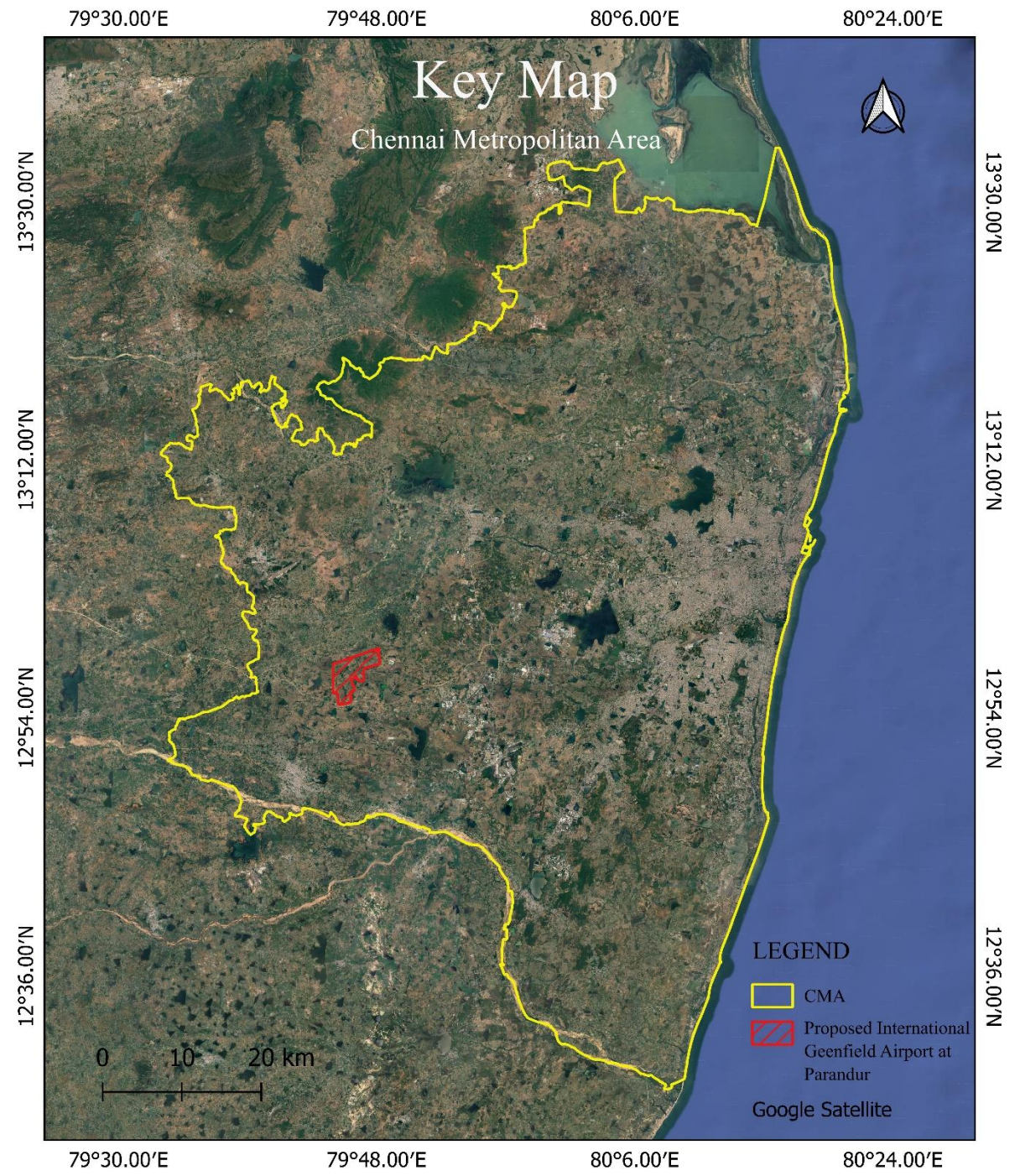
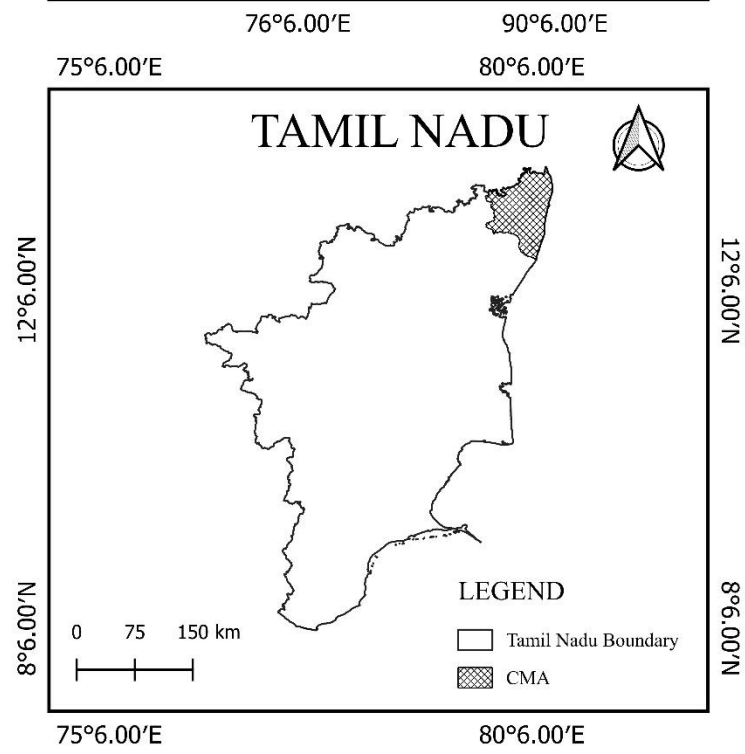
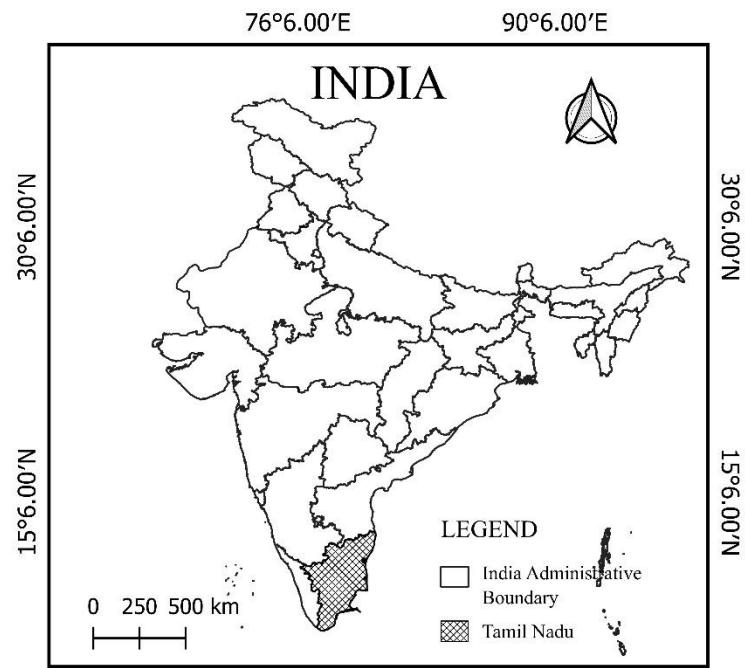


# STUDY AREA DESCRIPTION

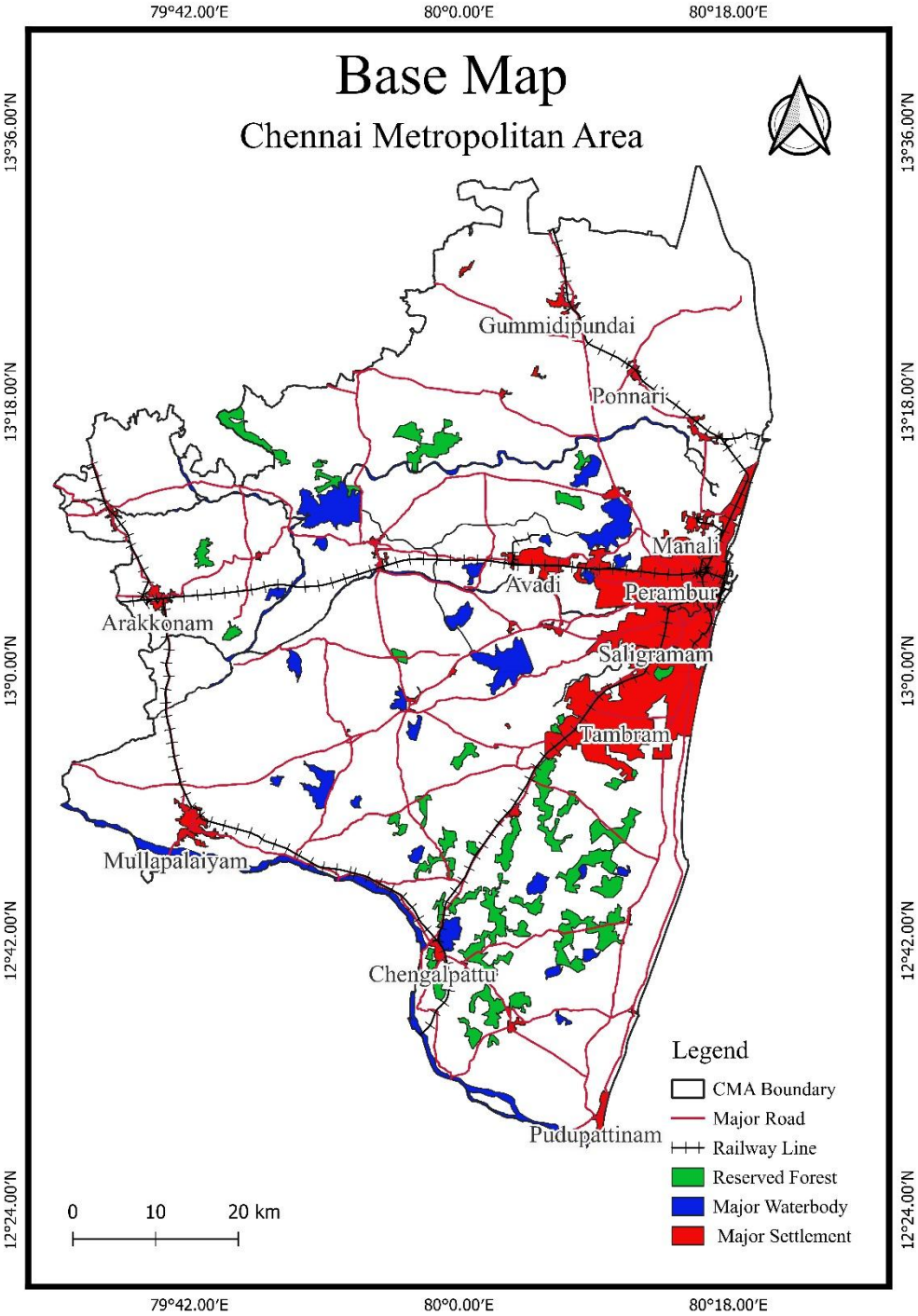
- The Proposed International Greenfield Airport at Parandur, located in the western part of the Chennai Metropolitan Area (CMA), Tamil Nadu, India. The airport site is situated near Parandur village in Kancheepuram district, approximately 70 km southwest of Chennai city.
- The Chennai Metropolitan Area (CMA), administered by the Chennai Metropolitan Development Authority (CMDA), was initially notified in 1974 with an area of 1,189 km<sup>2</sup>. In 2022, the Government of Tamil Nadu expanded the CMA boundary to **5,904 km<sup>2</sup>** to accommodate rapid urbanization and facilitate integrated regional planning.
- The expanded CMA encompasses Chennai city and parts of the Chengalpattu, Kancheepuram, Tiruvallur, and Ranipet districts, including the proposed Parandur Airport site.
- The study area is characterized by a heterogeneous landscape consisting of agricultural land, built-up areas, water bodies, plantations, barren land, transportation networks, and settlements. In recent decades, the Chennai has experienced rapid urban expansion driven by industrial development, improvements in transportation infrastructure, and increasing demand for residential and commercial land.

## **STUDY AREA DESCRIPTION** (*contd.*)

- The proposed airport is expected to further accelerate these land-use changes by attracting investments in housing, logistics, commercial establishments, industries, and supporting infrastructure.
- Because of these anticipated developments, the study area provides an appropriate setting for analysing historical land-use/land-cover (LULC) changes and simulating future urban expansion under different development scenarios. The results are expected to support sustainable land-use planning and informed decision-making within the Chennai Metropolitan Area.



# BASE MAP



# WORKS TO BE DONE

- Data collection (Landsat imagery, DEM, road network, ancillary data)
- Satellite image preprocessing
- Land Use/Land Cover (LULC) classification using Random Forest
- Accuracy assessment (Confusion Matrix & Kappa Coefficient)
- Preparation of urban growth driving factors
- Development of urban growth scenarios
- Hybrid CA–RF model implementation
- Future urban growth prediction
- Scenario-wise comparison and analysis
- Result interpretation and validation

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**THANK YOU**

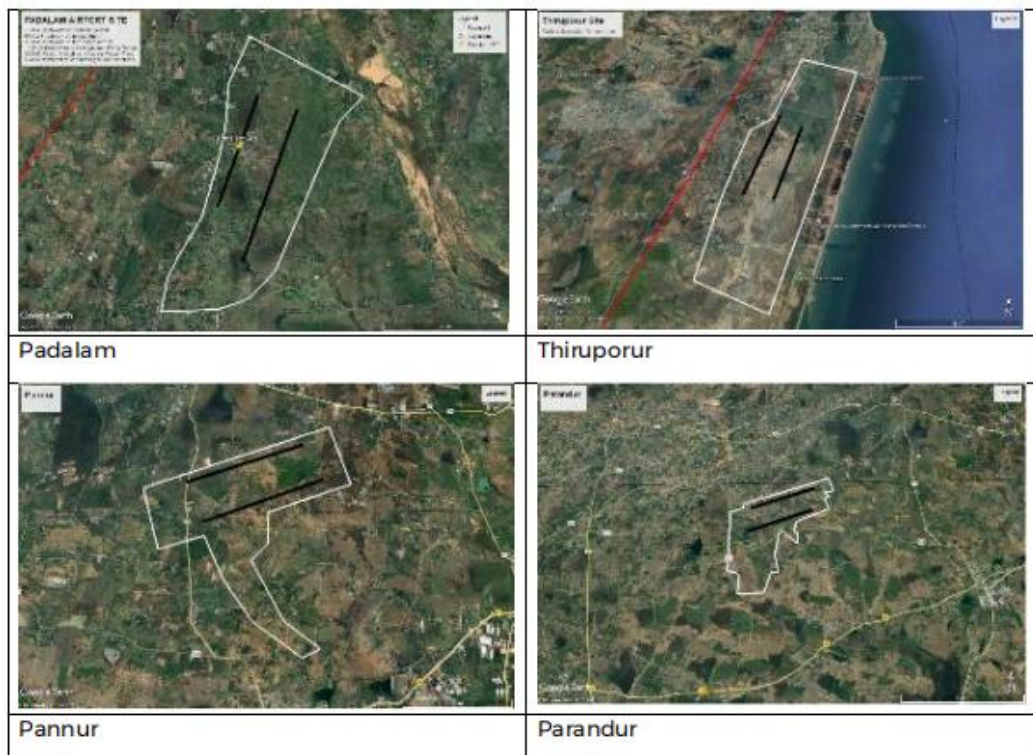
## JUSTIFICATION

TIDCO has conducted its basic level prefeasibility studies on identified potential sites. In order to take the process to next level, TN Government has asked AAI on 29.10.2021 to conduct technical prefeasibility study on those four sites. Airports Authority of India (AAI) had visited the above sites in February 2022 and conducted a Pre-Feasibility Study for the Proposed Greenfield Airport near Chennai.

The objective of this study was to evaluate potential four sites identified by Government of Tamil Nadu with respect to their suitability for Development of the Airport. The report was submitted by AAI in March 2022 documenting the findings of this study. The details of the four sites have been outlined below.

1. Parandur: The site is located northwest of Sungavarchatram.
2. Pannur: The site is located northeast of Sungavarchatram.
3. Thiruporur: The site is located southeast of Chennai (between ECR and RG Road).
4. Padalam: The site is located 20 kilometers south of Chengalpattu.

Figure: Analysis of Alternatives- Location Map



Source: AAI Pre-feasibility Report

The Pre-Feasibility study findings are that the Padalam site falls in the restricted area of Tambaram Local Flying area under the control of the Indian Air Force. The site at Thiruporur is adjacent to the restricted area of Tambaram Local Flying area. Hence there will be considerable restrictions on the flight operations (arrival and departures), thereby reducing the capacity of the proposed Airport. The Chengalpattu Firing range is present towards the North-East of Padalam site and West of Thiruporur site. Moreover, the prohibited area of Kalpakkam Nuclear plant is at a distance of 5NM to the South of the Thiruporur site.

The ideally suited runway orientation is 07-25 (parallel to Chennai Airport main runway). The site at Padalam and Thiruporur permits 02-20 orientation potentially conflicting with the traffic of existent Chennai Airport. Considering these factors and based on the available airspace, AAI has reported that Padalam and Thiruporur sites are not feasible for development of Greenfield airport.

The study by AAI concluded that the proposed sites in Parandur and Pannur are comparatively more feasible for development of Greenfield airport as per availability of airspace and being relatively free of natural obstructions. Both sites are free of any restrictions affecting operational efficiency and do not face any hazards and other limiting factors like wildlife sanctuaries, refuse dumps and sewage outfalls. Both sites have good connectivity to Chennai City and have adequate availability of ground access and trunk infrastructure. Land availability is adequate and suitable for the present requirement and for desired runway orientation of 07/25 at Parandur site. The land available at Pannur is 4284 acres and in Parandur is 5369 acres. However, additional land may be required for the Pannur site, to the East and Southeast side of the proposed area to accommodate essential navigational aids. Moreover, it was observed that several Extra High Tension (EHT) towers exist in and around the Pannur site. Moreover, another limiting factor of the Pannur site is that there are seven existing industries within and near the site. Based on the finding of the feasibility study carried out by AAI and further detailed analysis of both Parandur and Pannur sites, the Parandur site has emerged as the most suitable site for development of Greenfield Airport for the following reasons:

- Project site in Parandur is nearer to the upcoming Chennai - Bengaluru Expressway providing for easier and cost-effective connectivity to the demand centres, apart from other road & rail connectivity as explained above.
- Number of families to be displaced is 1005 in Parandur and 1546 families in Pannur site.
- There are lesser man-made obstacles for aircraft operations in Parandur compared to Pannur, where several EHT lines exist in and around the site and there are operating industries near the site.
- Sufficient land free of structures is available in the proposed site at Parandur for the operational and other needs of the Airport. Additional land on the East and Southeast side of the proposed area in Pannur may be needed to accommodate essential navigational aids. The additional land required in Pannur will be difficult to acquire as the land in the vicinity is developed with habitations and industries.
- Future development around the airport at Parandur can be planned better due to the availability of large tracts of vacant land, whereas the project area proposed in Pannur is near Sriperumbudur Industrial Area and is relatively developed with permanent industrial and residential structures.

In view of the above, after detailed consideration and technical study, Parandur was selected by the Government of Tamil Nadu for developing a new Greenfield Airport.



P.R.No.23/2025

Date: 11.03.2025

**Press Release**

**Detailed Project Report for Metro Rail Transit System between Poonamallee and Parandur**

Detailed Project Report (DPR) for the recommendation of the extension of Metro Rail between Poonamallee and Parandur was submitted by Thiru.M.A.Siddique, I.A.S., Principal Secretary to Government, Managing Director, CMRL to Dr.K.Gopal I.A.S., Additional Chief Secretary to Government, Special Initiatives, Government of Tamil Nadu at Secretariat, Chennai today (11.03.2025).

Thiru. T. Archunan, Director (Projects), Thiru. S. Krishnamoorthy, Director (Finance), Thiru. T. Livingstone Eliazer, Chief General Manager (PP & D), Thiru.R.Narendrakumar, JGM (P&BD) and other Senior Officials of CMRL were present during the occasion.

The proposed Metro rail extension is starting from Poonamallee, passing through Chembambakkam, Thandalam, Irungattukottai, Sriperumbudur, Sunguvarchatram and ending at Parandur Airport, which will provide seamless connectivity to the Kuthambakkam Bus Terminus and proposed green field Airport. It is integrated with the proposed Elevated Corridor by NHAI between Poonamallee and Sriperumbudur for a length of 5.892 Km to optimize the project cost and implementation time. The project details are as follows.

- Total Length Corridor: 52.94 km.
- No. of Elevated Stations: 20 No.
- Estimated Completion cost: INR 15,906 Crores.

As part of project implementation, the consultant has recommended implementing from Poonamallee to Sunguvarchatram under Phase-1 and the details of the same are as follows.

- Total Length Corridor: 27.90 km.
- No. of Elevated Stations: 14 No.
- Estimated Completion cost: INR 8,779 Crores.

(The above estimated cost excludes NHAI elevated corridor integration cost.)

**Issued By: Joint Director / Public Relations Officer**  
CMRL, Chennai – 600 035.

GOVERNMENT OF INDIA  
MINISTRY OF CIVIL AVIATION  
**RAJYA SABHA**  
**UNSTARRED QUESTION NO. : 38**  
TO BE ANSWERED ON THE 20th July 2026  
**PARANDUR AIRPORT IN TAMIL NADU**

38. SHRI P. WILSON

Will the Minister of CIVIL AVIATION be pleased to state:-

- (a) the current progress of the proposed Greenfield airport at Parandur, Chennai, including stage-wise progress, total estimated project cost, budgetary allocation made so far and the anticipated timeline for completion;
- (b) whether the new Tamil Nadu Government has requested the Central Government to change the proposed site of the Greenfield Airport, if so, the details of such request, date-wise, and the response of Government thereon; and
- (c) the present status of expansion and development works at Chennai, Coimbatore, Salem, Tiruchirappalli, Madurai and Tuticorin airports, indicating the project-wise budget allocation, physical and financial progress, expected date of completion and other relevant details?

**ANSWER**

Minister of State in the Ministry of CIVIL AVIATION (Shri Murlidhar Mohol)

(a): Ministry of Civil Aviation (MoCA) has granted 'In-Principle' approval to TamilNadu Industrial Development Corporation Limited (TIDCO), a Government of Tamil Nadu Enterprise for development of an International Greenfield Airport at Parandur on 07.04.2025. As per In-Principle application submitted by TIDCO in September, 2024, estimated project cost for Phase-1 is approximately Rs. 10500 crore.

As per the Greenfield Airports Policy, 2008, the responsibility for implementation of the Greenfield airport projects including land acquisition, R&R, funding, obtaining environment clearances etc. lies with the airport developer concerned or the State Government, as the case may be. The timeline for completion of the airport projects depends upon many factors such as land acquisition, availability of mandatory clearances, financial closure etc.

(b) No.

(c): Upgradation/modernisation of existing airports is a continuous process. Major expansion and development works at Chennai, Tiruchirappalli and Coimbatore airports include:

# Airport uncertainty won't affect metro rail extension

U Tejonmayam / Jul 28, 2026, 00:03 IST

Preferred on 

 Comments

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AA

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**Chennai:** The proposed 52.94km metro rail extension from Poonamallee to Parandur has not been shelved despite uncertainty over the proposed greenfield airport. Chennai Metro Rail Ltd (CMRL) is moving ahead with approvals for the first phase from Poonamallee to Sunguvarchatram, covering 27.9km.

The move comes even as the Parandur airport project remains in limbo. While the Union govt said it has not received any proposal from Tamil Nadu to shift the

airport site, state officials have indicated that land acquisition has been put on hold and the project is under review, although the Centre's in-principle approval remains valid.

"The metro extension project is not dropped. The approval for the line up to Sunguvarchatram is in process," a CMRL official said.

The official said the Centre is processing all three proposed metro extensions — Airport-Kilambakkam, Koyambedu-Avadi and Poonamallee-Sunguvarchatram. "They are very positive on these three extension lines," the official said.

The original 52.94km elevated Poonamallee-Parandur corridor with 20 stations was estimated to cost ₹15,906 crore. The project was conceived in phases, with Poonamallee-Sunguvarchatram to be taken up in phase-1 and Sunguvarchatram-Parandur stretch to be taken up later in line with the airport's development.

The state govt had accorded in-principle approval and earmarked ₹2,126 crore in Sept 2025 for preparatory work on the first phase, including ₹1,836 crore for land acquisition.

The Centre's approval is now being sought for the ₹8,779-crore Poonamallee-Sunguvarchatram section, which will have 14 elevated stations. The corridor is expected to serve fast-growing residential areas, the Sriperumbudur industrial belt and the proposed Kuthambakkam bus terminus. Officials earlier said trains could operate at 80kmph on the corridor and be accelerated beyond 100kmph on the future Sunguvarchatram-Parandur stretch.

Meanwhile, CMRL is awaiting a date from the Centre to inaugurate the first operational stretch of Phase II between Poonamallee Bypass and Vadapalani. The corridor will connect with the existing Phase I network at Vadapalani through an interchange station, providing seamless connectivity for commuters.

"Even if the airport gets delayed, the metro line up to Sunguvarchatram should go ahead because we need an alternative and faster transportation to the city. Also, Sriperumbudur and Sunguvarchatram witness heavy traffic, and thousands of employees commute to the city daily. This corridor is needed regardless of what happens to the airport," said R Indumathy, a resident of Sriperumbudur.